

Mid-Term Presentation

Influence Maximization

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Problem Details

Community-based Influence Maximization in Attributed Networks

Core Question

Which users to incentivize so adoption spreads maximally within budget and across the right communities?

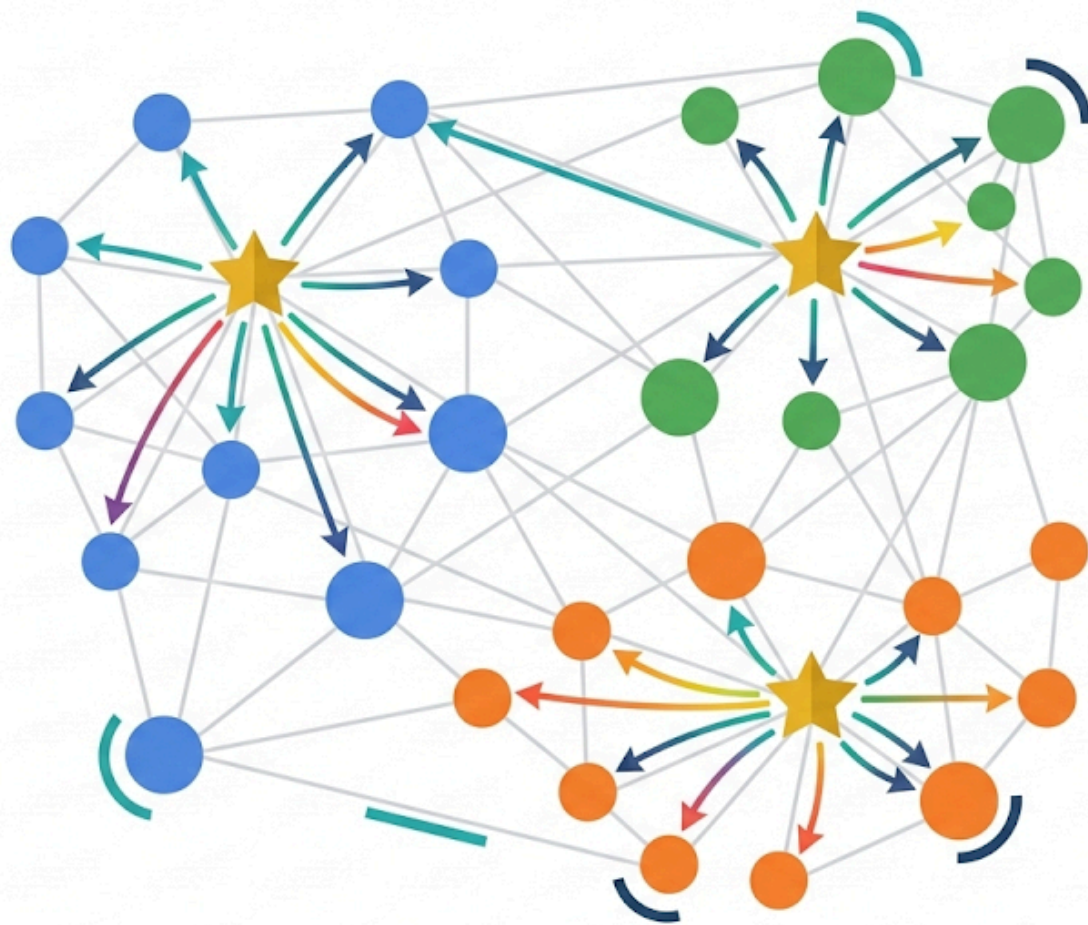


Fig: Influence propagates from seeds (★) through the network, reaching different communities

BASE PROBLEM

Influence Maximization (IM)

Select k seed nodes in a social network to maximize information spread under the Independent Cascade model. Classical greedy (CELF) and sampling-based (IMM) methods are too slow or inflexible at scale.

EXTENSION 1

Budget Constraints

Each seed node has a cost (e.g., Influencer incentive fee). Total spend must stay within a fixed campaign budget

EXTENSION 2

Group-Sensitive Objectives

Nodes belong to structural communities. The campaign must control how influence distributes across segments balanced reach across all groups

RL Formulation

Problem is sequential, budget-limited, depends on graph structure, has stochastic environment and requires reward maximisation

STATE

● Graph topology

Graph structure and already influenced nodes

● Seed set so far

Which nodes have already been selected as seeds in the current episode

● Remaining budget

How much of the campaign spend limit is still available

● Formally:

$S_t = (\text{Graph } G, \text{Selected set, Remaining Budget})$

ACTION

● Select one node

Select one unchosen node

● Budget

Node cost must fit remaining budget

● Large but discrete

Thousands of possible actions per step RL avoids exhaustive combinatorial search

● Marginal gains

Sequential selection naturally captures diminishing returns of each additional seed

REWARD

● Marginal spread

Reward = Marginal increase in influence spread

● Community shaping

Reward term reflecting how well influence distributes across target segments

● Total reward

Total episode return = Total influence spread

● Episode Ends when Budget exhausted

CELF — Cost Effective Lazy Forward

About the Dataset

Full Graph

~4M

Nodes

34.6M

Edges

LiveJournal — a real blogging platform where users follow each other and form interest groups

What We Use

Top 5,000 Communities File

139,012 unique nodes covered

3 – 1,441 community size range

27.8 average community size

1.6 avg memberships per node

~1,000 nodes in our working subgraph (top 10 communities)

CELF — How We Pick Seeds

- 1 **Run IC for every node alone**
score = influence reach
- 2 **Push all scores into a max-heap**
best influencer always at top
- 3 **Pop top → re-evaluate with current seed set**
accept if still best
- 4 **Repeat until budget is exhausted**

Max-Heap

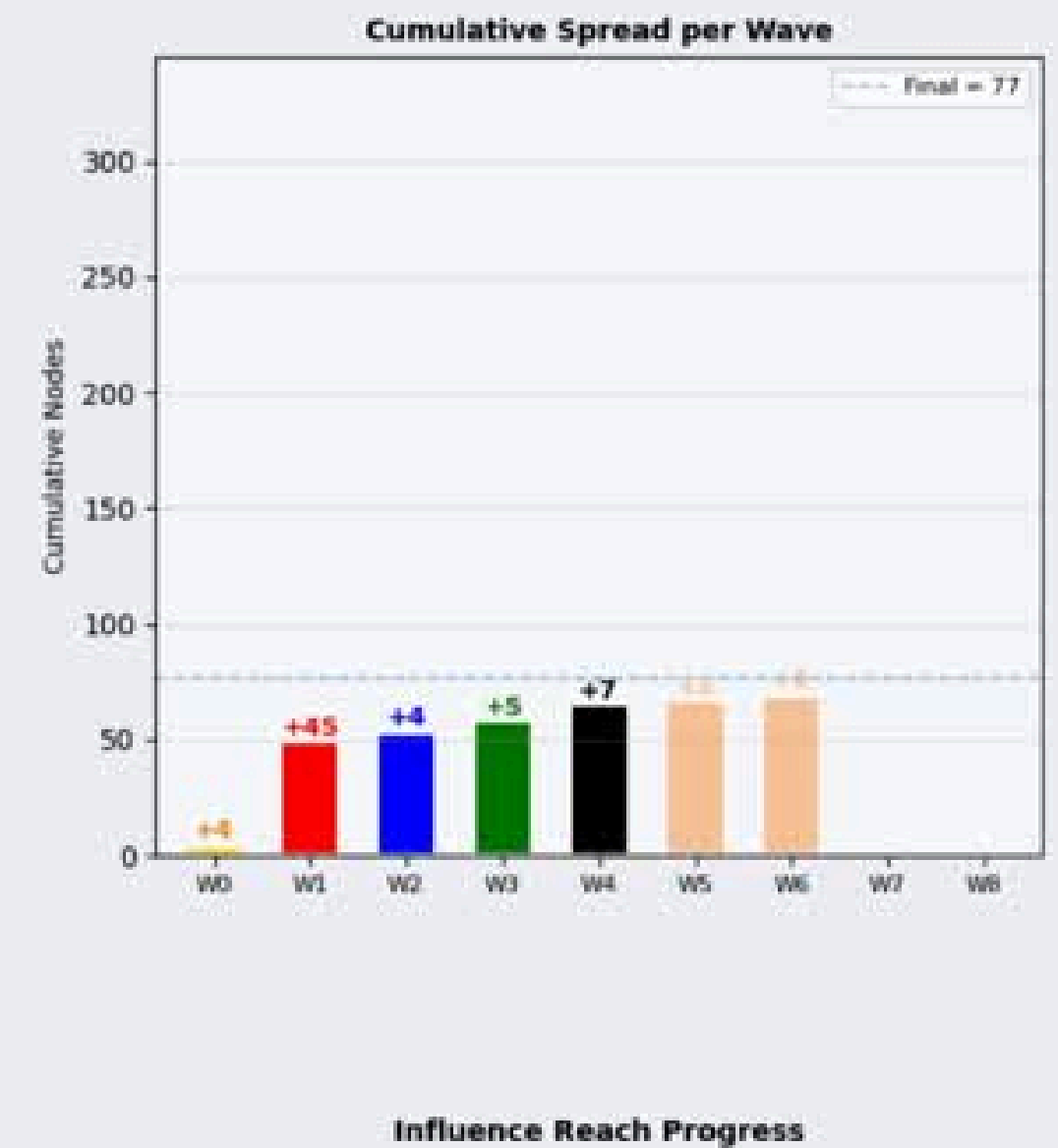
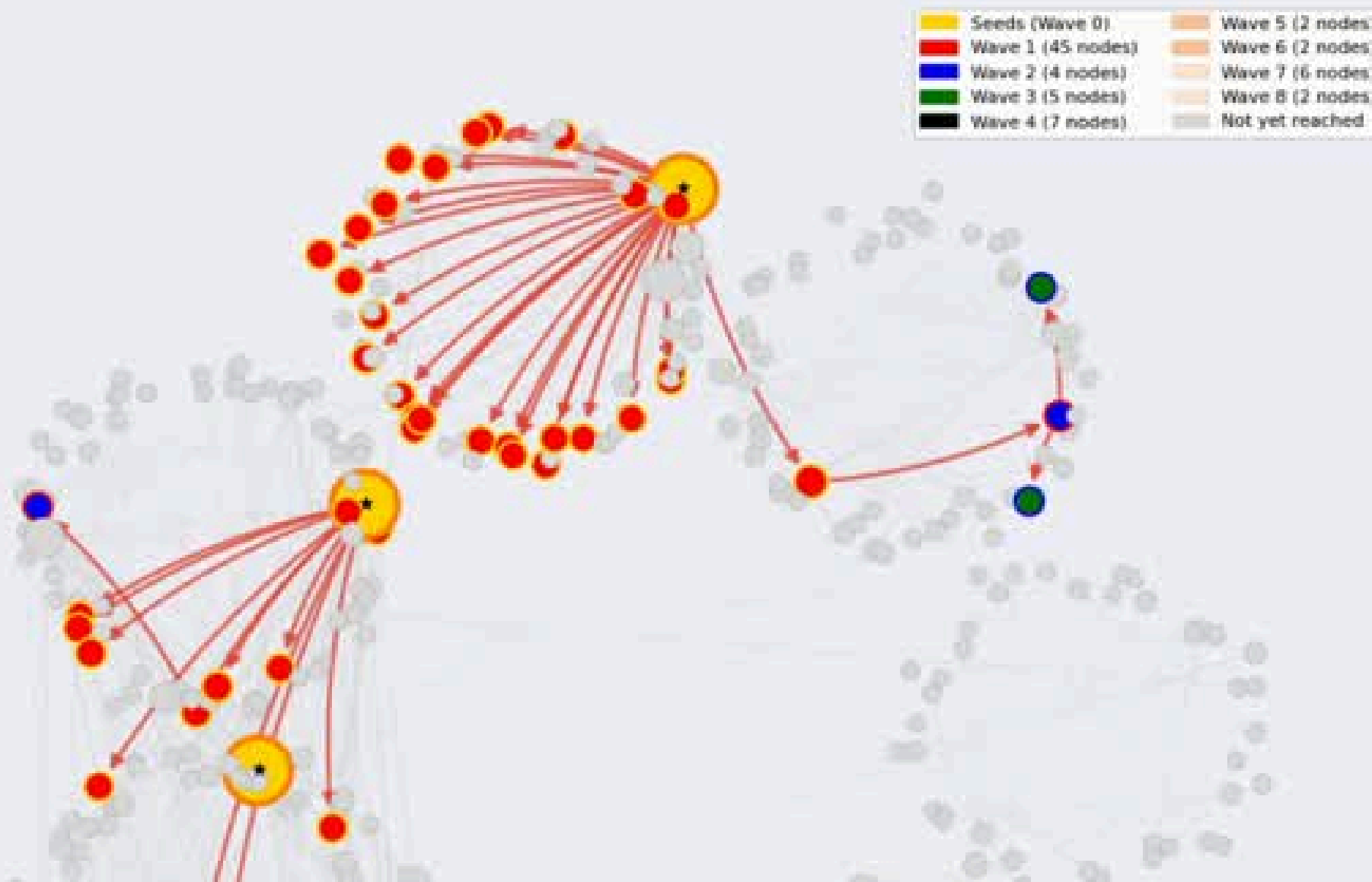
▼ Pop from here

Alice	score: 47
Bob	score: 38
Carol	score: 35
Dave	score: 29
Eve	score: 12

Limitations

- Slow — scores every node before picking anyone
- Re-checks expensive — 150 IC simulations each time
- No community awareness

[CELF] IC Spread Animation
Wave 6 / 8 — 69 / 493 nodes activated (14.0%)



IMM — Influence Maximization via Martingales

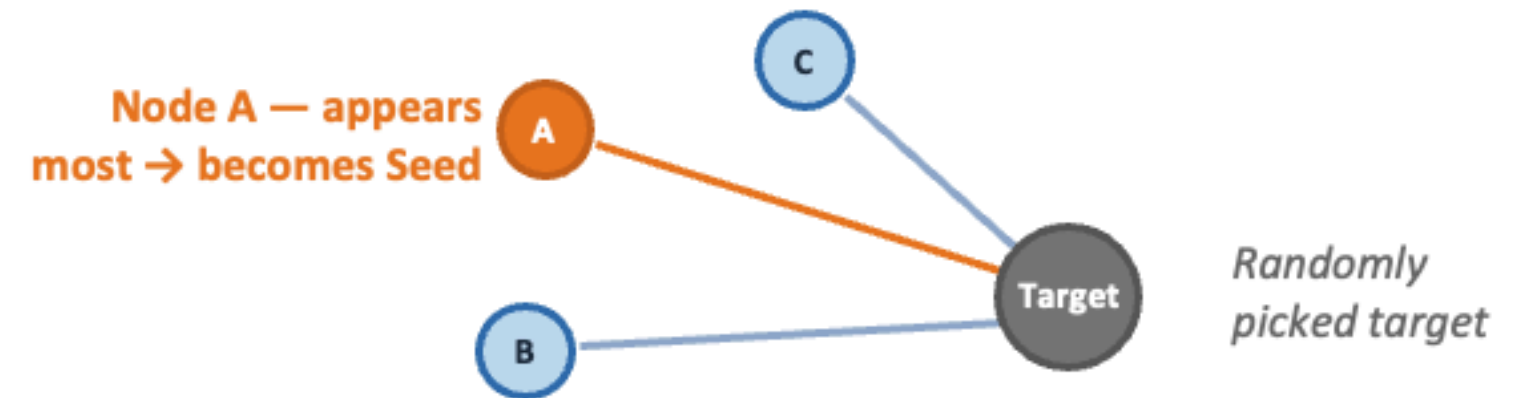
IMM — How We Pick Seeds

- 1 Estimate max affordable seeds k from budget
- 2 Pick a random node → trace backwards through graph
this is one Reverse Reachable (RR) set
- 3 Repeat thousands of times
build a coverage index — which nodes appear in which RR sets
- 4 Greedily pick node covering most new RR sets per unit cost
repeat until budget exhausted

Limitations

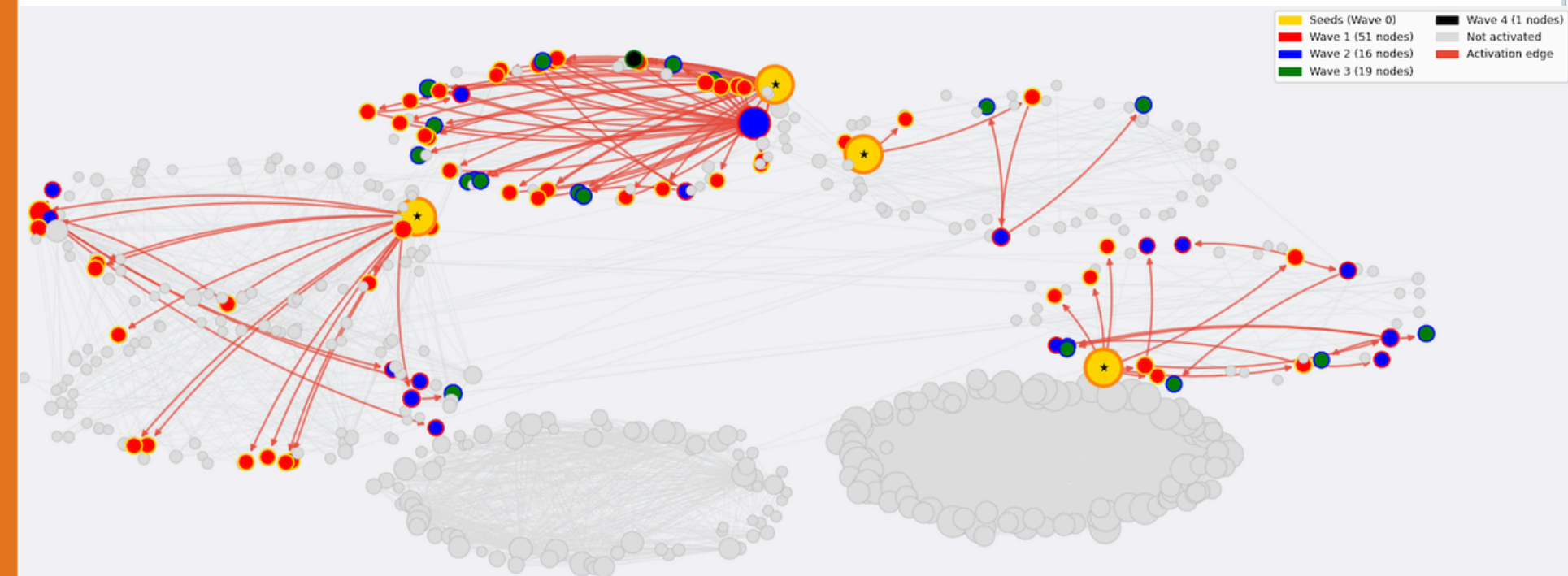
- RR set quality depends on graph structure
- No community awareness — same gap as CELF

What is a Reverse Reachable (RR) Set?



"Who could have reached this person?" — repeat thousands of times → most frequent node = best seed

IMM in Action — IC Wave Propagation



Seeds=4 | Waves=4 | Activated=91/493 nodes | Expected Spread: 81.2 ± 17.8 nodes

ToupleGDD — Architecture Overview

Three Coupled GNNs + Double Deep Q-Networks for end-to-end IM learning

STEP 1

Personalized DeepWalk (PDW)

Goal: Learn initial node embeddings capturing both local & global influence context

Local Context: We use Random Walk with Restart (RWR) to sample nodes that you can activate

Global Context: Uniform sample from r-hop out-neighbors

Output: X_u (state), S_u (source), T_u (target)

STEP 2

Three Coupled GNNs (ToupleGNN)

State GNN: Models activation probability X_v — aggregates from in-neighbors

Source GNN: Models capacity S_v to activate others — aggregates from out-neighbors

Target GNN: Models tendency T_v to be influenced — aggregates

Complexity: K layers, $O(|V| + |E|)$ per layer

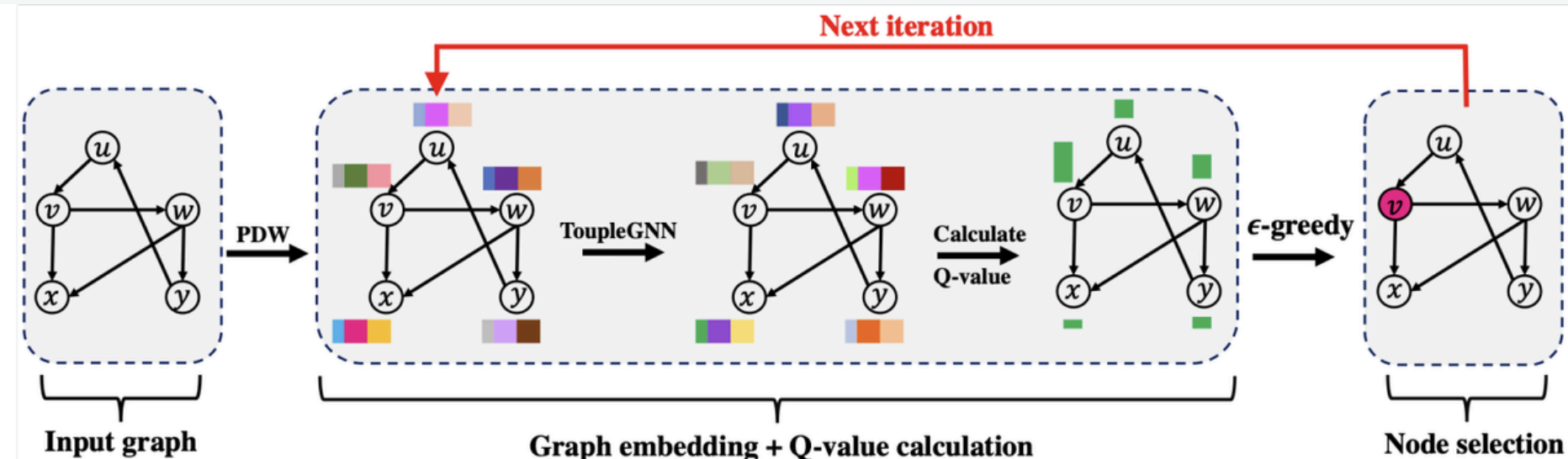
STEP 3

RL via Double DQN (DDQN)

Q-Function: $\hat{Q}(u, S_t; \Theta)$ scores marginal gain using concatenated $[X_u, S_u, T_u]$

Behavior + Target Networks: Avoids over-optimistic Q estimates of plain DQN

Training: ϵ -greedy + n-step Q-learning + experience replay for stable convergence



Timeline

